Peter Sitati

Technology Leader · AI, Fintech & Platform Engineering

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SUMMARY

Innovative technology leader with 15+ years building scalable software platforms, leading cross-functional teams, and driving digital transformation across fintech, media, agritech, and infrastructure. Proven track record shipping production AI systems including voice-driven applications, multi-channel agentic platforms, LLM-powered automation, financial reconciliation and data-migration pipelines, alongside mobile apps, payment systems, and enterprise integrations.

PROFESSIONAL EXPERIENCE

Chief Executive Officer · **NARS Technical Services** *Mar 2022 to Present · Nairobi, Kenya*

- Designed and shipped a production **LLM-as-judge evaluation system** (DSPy) for multi-tier grading of open-ended learner answers with rubric dimensions, confidence scoring and human-review escalation, inside a **voice-first AI learning platform** (LiveKit/WebRTC, LangGraph, Deepgram, ElevenLabs) with provider failover across OpenAI/Google/Soniox, subscription billing, and 2,300+ automated tests on Fly.io, with grading and speech-to-text accuracy ratcheted into CI.
- Architected a **multi-channel AI lead-nurturing engine** (LangGraph + FastAPI) coordinating voice (Retell AI), WhatsApp, and SMS with Claude conversational logic, a Qdrant **RAG** knowledge base, and **M-Pesa**-integrated multi-tenant billing, serving multiple enterprise clients.
- Developed a **settlement reconciliation platform** (FastAPI, React, LangGraph) automating extraction of bank settlement data across **180 accounts in 38 African countries**: NIBSS, KEPSS, PAPSS, SWIFT.
- Engineered an **enterprise data-migration pipeline** ingesting ~12,000 legacy financial documents into ERPNext via a 4-stage parser chain with LLM fallback (Claude/Cerebras) and vector-embedding semantic deduplication (ChromaDB).
- Created an **AI content management agent** (Mastra, Claude) to autonomously grade, optimize SEO/GEO, and curate **1,200+ blog posts** for an EV publication focused on developing economies.
- Built **Run.KE**, a running events platform for Kenya (Wasp 0.21, React, Node.js, Prisma): event discovery and admin tooling with **LLM-vision flyer parsing** for stylized posters, running-track and gym directories (OpenStreetMap), a route tool, Cloudflare R2 presigned uploads, Turnstile bot protection, and scheduled analytics jobs.
- Built an **agentic content manager** (Mastra, Claude) for a health publication that grades drafts across mission-alignment and technical-quality dimensions, optimizes SEO, and publishes to Ghost CMS under YMYL/E-E-A-T standards.
- Integrated **gbrain**, a local-first **knowledge-graph memory system** served over **MCP**, across my most important repositories as a working **knowledge base**, writing the ingestion, rescan, remote-sync and health-check tooling around it and the query playbooks it is used through: semantic retrieval over local Ollama embeddings in PGLite, graph traversal, entity/fact extraction, and contradiction detection over the corpus.
- Built a **tender document automation tool** (Python, PyMuPDF, Claude API) and **eTIMS integration tooling** for KRA tax-invoice compliance (20+ endpoints); engineered a DeFi arbitrage trading bot (Solidity, Web3.py).
- Led deployment of secure, high-availability networking infrastructure (fiber, wireless, LAN) for enterprise clients.

Head of Mobile · **TradePulse Inc.** *Jan 2024 to Aug 2025 · Nairobi, Kenya*

- Led end-to-end Flutter development of the **TradePulse Mobile App**, enabling agro-processors to manage inventory, create orders, accept payments, and generate reports.
- Spearheaded React/Next.js development of the **TradePulse Sourcing Platform**, a CRM-driven B2B system streamlining raw-material sourcing between buyers and suppliers; scaled regionally to hundreds of daily transactions.

Team Leader, Innovation / Software Engineer · **Radio Africa Group** *May 2016 to Apr 2022 · Nairobi, Kenya*

- Directed agile teams building consumer digital products reaching **millions of users**: a music streaming platform with real-time recommendations, a digital newspaper with subscriptions, and interactive games/quizzes.
- Championed modern software practices and cloud infrastructure to accelerate development cycles and digital transformation.

Software Engineer · **Cellulant Kenya** *Nov 2011 to Apr 2016 · Nairobi, Kenya*

- Developed and maintained **bulk payment systems** for enterprise and government clients, enabling high-volume secure transactions.
- Built payment systems powering the **Growth Enhancement Support Program** (Nigeria), connecting input suppliers to **millions of farmers** to deliver subsidies efficiently.

Bioinformatician · **IITA** · **Earlier roles** *2005 to 2011*

- Built data-mining and bioinformatics workflows for genomic data analysis at IITA (2010 to 2011). Prior: Research Assistant, Innovative Health Research · Network Installations Coordinator, NARS · Drupal Integrator, Koumbit Network (Montreal) · Web Developer, Norpix Inc. & Canadian Chemical Biology Network.

SELECTED PROJECTS

- **Quizlar**: Voice-first AI learning platform with real-time WebRTC quiz sessions and FSRS spaced repetition, exposed to external agents through an MCP server. LiveKit · LangGraph · DSPy · Deepgram · FastAPI · Flutter · React · PostgreSQL
- **Multi-Channel Voice Agent Engine**: Agentic voice/WhatsApp/SMS lead-nurturing platform with RAG and M-Pesa billing. Retell AI · Claude · Qdrant · MySQL + PostgreSQL · React/TypeScript
- **TradePulse Sourcing Platform**, B2B supply-chain CRM for agricultural sourcing: role-based dashboards for administrators, agro-processors and suppliers, covering aggregated demand, supplier records, quote lifecycle, deals, orders, payments and wallets. Next.js 15 · React · TypeScript
- **Doc Sift**: Document extraction pipeline turning invoices, POs, receipts and bank statements into structured records synced to the client's system of record, with an eval harness on hand-checked ground truth. TypeScript · LLM extraction
- **Settlement Reconciliation Platform**: 15-stage pipeline normalising ten settlement formats (M-PESA, MTN, Airtel, NIBSS, KEPSS, PAPSS, SWIFT MT940/MT103, ISO 20022 camt.053) across ~180 accounts in 38 African countries. Python · FastAPI · React · LangGraph
- **ERPNext Accounting Ingestion**: 12k-document financial migration with LLM-fallback parsing and vector dedup. Python · ChromaDB · Docling · Claude/Cerebras · ERPNext REST API
- **Run.KE**, Running events platform for Kenya: event discovery/admin with LLM-vision flyer parsing, track & gym directories, route tool. Wasp 0.21 · React/TypeScript · Node.js · Prisma · Cloudflare R2 + Turnstile
- **Healthspan Content Agent**: Agentic content manager that grades drafts on mission-alignment and technical quality, optimizes SEO, and publishes to Ghost CMS. Mastra · Claude API · Ghost CMS · TypeScript
- **Cross-repository knowledge base**: gbrain, a local-first MCP knowledge-graph memory system, integrated across my most important repositories, with the ingestion, rescan, sync and health-check tooling written around it. gbrain · PGLite · Ollama embeddings · MCP · Claude API
- **Plopped**, Bristol-scale bowel health diary, shipped solo to iOS: Premium subscription (in-app purchase), engagement-gated ads, hands-free voice logging (on-device STT into LLM field extraction, reviewed before save), biometric lock, timezone-aware reminders, offline-first local store. Documented domain model and ADRs; CI-built releases. Flutter · Riverpod · Drift (SQLite) · Firebase Analytics · AdMob · StoreKit

TECHNICAL SKILLS

Languages & Frameworks: Flutter, FlutterFlow, React, Next.js, Node.js, Wasp (Open SaaS), FastAPI, JavaScript, TypeScript, Python, PHP, Solidity, SQL, Bash

AI & Voice: LLM Evaluation & Evals (LLM-as-judge), LangGraph, DSPy, Mastra, n8n, Claude API, OpenAI, MCP (built a FastMCP server), Qdrant (RAG), ChromaDB, PGLite, Ollama, Deepgram (STT), ElevenLabs (TTS), LiveKit (WebRTC), Retell AI

Data & Platforms: PostgreSQL, MySQL, SQLite, Redis, AWS, GCP, Fly.io, Docker, n8n, Ghost CMS, CI/CD, REST APIs

Other: LLM Evaluation Design, AI Agent Orchestration, RAG Systems, Data Extraction Pipelines, Systems Architecture, M-Pesa Integration, Agile Leadership

EDUCATION

Jomo Kenyatta University of Agriculture & Technology
BSc Biochemistry & Molecular Biology

McGill University
Biochemistry & Molecular Biology coursework, credits transferred to JKUAT

MORE

Languages: Swahili (Native), English (Native), French (Working)

Certification: Java Developer Certificate

Publication: *Genomics & Potential Downstream Applications in the Developing World*